

APPLICATIONS SECTION

A DYNAMIC LINEAR MODEL FOR PRICING FIXED INTEREST RATE STOCKS ON THE SOUTH AFRICAN MARKET

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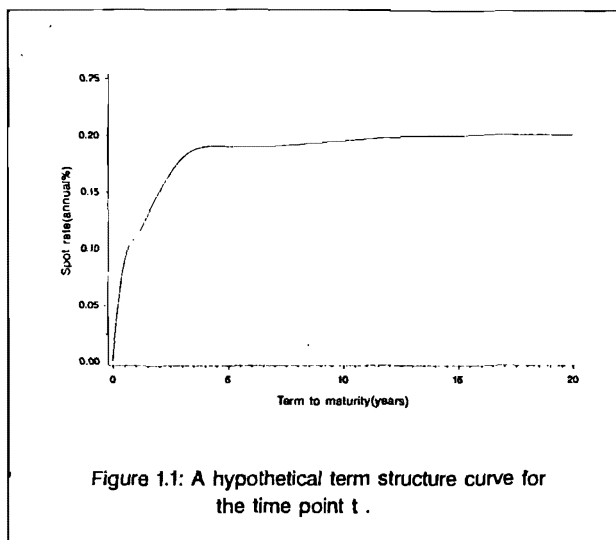
Summary: A study of the South African financial markets is undertaken with a view to modelling the underlying term structure of interest rates that affect investment opportunities. A dynamic linear model will be developed with the measurement equation being used to model the observed fixed interest rate stock prices that occur on the market, and the system equation being used to model the unobservable spot rates of interest that prevail on the market.

1. INTRODUCTION

In an economic setting, a spot rate of interest measures the rate of exchange that exists between money that is exchanged today for money at a single date in the future. For example, if a single payment Y_t , due to be made in Δ years time, has a current (time t) value of X_t , then the spot rate of interest prevailing at this current point in time and which is pertinent for a term to maturity period of Δ years, is defined as the rate of exchange that exists between X_t and Y_t . Using the notation $R(t, \Delta)$ to denote the current spot rate of interest (where the first argument t denotes the current time point and the second argument Δ the term to maturity on the loan) we have

$$X_t(1 + R(t, \Delta))^\Delta = Y_t$$

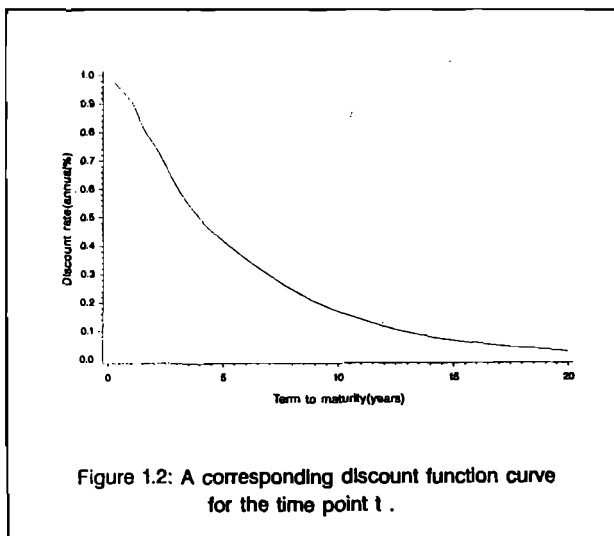
defining the relationship between X_t and Y_t . By considering similar types of payment schemes that prevail at time t , but which have differing terms to maturity, a plot of the resulting spot rates against their terms to maturity characterize a specific pattern of interest rates that is called the term structure for that time point t (see figure 1.1). It is the aim of this paper to model this term structure curve as the current time index t moves forward in time.



Given the above set of definitions

$$(1.1) \quad D(t, \Delta) = (1 + R(t, \Delta))^{-\Delta}$$

can be viewed as representing a discount factor rate, prevailing at the time point t , for a maturity period of Δ years. When plotted against their term to maturity these discount factor rates define a discount function curve for the time point t that is similar to the one given in figure 1.2.



In South Africa there exists an active market for the buying and selling of securities that pay a half-yearly fixed rate of interest, known as the coupon rate, over a period of time that is terminated by a date of redemption that is printed on the stock. On this date, a principal value is then also paid to the stockholder, to redeem the value of the stock. These securities then provide a database from which one can attempt to estimate the term structure curve that currently prevails in the market.

Letting

N = the nominal or face value of the stock

C = the annual coupon rate expressed as a percentage of the stock's nominal value

n = the number of coupon payments still to be made by the stock

δ_i = the fraction of time remaining (in years) until the i -th coupon payment is to be made

the application of present value arguments suggests that the following pricing equation can be used for extracting the currently prevailing discount function rates from a set of coupon bearing securities, viz.

$$(1.2) \quad P(C, t, n) = (NC/200) \sum_{i=1}^n D(t, \delta_i) + ND(t, \delta_n)$$

where $D(t, \delta_i)$ denotes the current (time t) discount factor rate pertaining to a maturity period of δ_i years and $P(C, t, n)$ the current price of the stock.

In the next section we will attempt to model the above pricing equation in the form of a state space model where the discount function curve that is represented for the time point t in figure 1.2 is approximated by a piecewise linear function with relatively few parameters. To this end, let us define the state space model representation that we will be using as follows:

$$\mathbf{y} = \mathbf{F}_t \mathbf{x}_t + \mathbf{v}_t$$

$$\mathbf{x}_t = \mathbf{G}_t \mathbf{x}_{t-1} + \mathbf{w}_t$$

$$\mathbf{x}_0 \sim N(\hat{\mathbf{x}}_{0|0}, \Sigma_{0|0})$$

where

$\mathbf{y}_t = n_t \times 1$ vector of process observations that have been collected at the time point t

$\mathbf{x}_t = p \times 1$ random vector of unobservable process parameters (state variables)

$\mathbf{F}_t = n_t \times p$ measurement equation matrix assumed to be known at the time point t

$\mathbf{G}_t = p \times p$ system equation matrix assumed to be known at the time point t

$\{\mathbf{v}_t\}$ = a sequence of uncorrelated random vectors with zero mean and variance covariance matrix $E\{\mathbf{v}_t \mathbf{v}_t'\} = \mathbf{R}_t$

$\{\mathbf{w}_t\}$ = a sequence of uncorrelated random vectors with zero mean and variance covariance matrix $E\{\mathbf{w}_t \mathbf{w}_t'\} = \mathbf{Q}_t$

The model description is then completed by assuming that

$$\begin{aligned} E\{\mathbf{w}_t \mathbf{w}_s'\} &= 0 & E\{\mathbf{v}_t \mathbf{v}_s'\} &= 0 & E\{\mathbf{w}_t \mathbf{v}_s'\} &= 0 \\ E\{\mathbf{w}_t \mathbf{v}_t'\} &= 0 & & & & \text{for } t \neq s \end{aligned}$$

and that the initial state parameter vector, $\mathbf{x}_0$, is uncorrelated with $\{\mathbf{v}_t\}$ and $\{\mathbf{w}_t\}$.

The EM algorithm as developed by Shumway and Stoffer (1982), can then be used to estimate the unknown model parameters in the above state space model with the Kalman smoothing equations then being used to obtain suitable parameter estimates for the state parameter vector, viz.

$$\begin{aligned} \hat{\mathbf{x}}_{t|T} &= E\{\mathbf{x}_t | \mathbf{y}_1, \dots, \mathbf{y}_T, \hat{\boldsymbol{\theta}}\} \\ \Sigma_{t|T} &= E\{(\mathbf{x}_t - \hat{\mathbf{x}}_{t|T})(\mathbf{x}_t - \hat{\mathbf{x}}_{t|T})' | \mathbf{y}_1, \dots, \mathbf{y}_T, \hat{\boldsymbol{\theta}}\} \end{aligned}$$

where $\hat{\boldsymbol{\theta}}$ denotes the maximum likelihood estimate that the EM algorithm has provided for the unknown model parameters.

2. THE DEVELOPMENT OF THE MODEL

Let us assume that we have been collecting a sample of coupon bearing stocks over the time period points $t = 1 \dots T$. Employing the notation

$$\mathbf{y}_t = (y_t^{(1)}, \dots, y_t^{(n_t)})'$$

to denote the vector of stock price observations that have been collected at the time point t , a term structure based model for pricing these observations can, in view of (1.2), be given by

$$(2.1) \quad y_t^{(i)} = (N^{(i)} C^{(i)} / 200) \sum_{j=1}^{m_i} D(t, \delta_j^{(i)}) + N^{(i)} D(t, \delta_{m_i}^{(i)}) + e_t^{(i)}$$

where

$y_t^{(i)}$ = the market price of stock i at the time period point t

$C^{(i)}$ = the coupon rate of stock i expressed as an annual percentage of the stock's nominal value

$N^{(i)}$ = the redemption value of stock i

m_i = the number of coupon payments still to be made by stock i

$\delta_j^{(i)}$ = the fraction of time remaining (in years) until the j -th coupon payment of stock i is due to be made

$D(t, \delta_j^{(i)})$ = the appropriate discount factor pertaining to a payment of R1 in $\delta_j^{(i)}$ years time

$e_t^{(i)}$ = a random error term that is assumed to capture any deviations that may occur between the theoretical and observed price of stock i

$i = 1, \dots, n_t$ indicating that the sample size of stocks that are collected for each time period point t may vary

From the above pricing equation, however, it becomes apparent that an identification problem will arise in that the number of discount factor rate parameters that will need to be estimated in order to make the above model operational for the time period point t will generally far exceed the number of stock price observations that have been collected.

If in fact all the coupon payments, which are usually paid on a bi-annual basis, were to take place at the same point in time, then we would require that

$$n_t \geq \max\{m_1, \dots, m_{n_t}\}$$

in order to make the estimation problem feasible. Clearly the problem lies in trying to estimate a whole sequence of discount factor rate points on the

discount function curve rather than attempting to estimate a few model parameters that can be used to describe the discount function curve itself.

In order to reduce the dimensionality of the above estimation problem to a more manageable proportion, the use of classical approximation theory would suggest that one tries to approximate the discount function, prevailing at the time point t , by a linear combination of suitably chosen approximating functions. The net result of this would be the replacement of, what would be the estimation of a large number of unknown discount factor points, by a significantly smaller number of unknown parameter coefficients, namely those that have been employed in the approximation.

Schaeffer (1973) developed such a model using Bernstein polynomials to approximate the discount function. Arguing that the discount function is essentially exponential in shape, Vasicek and Fong (1982) developed a model that used exponential spline functions. McCulloch (1975) chose to use quadratic and cubic spline polynomial approximations. All the above models have, however, tended to concentrate on the development of a static model of the term structure, static in the sense that a term structure curve was estimated only for a specific point in time, i.e. no attempt was made to link the term structure of yesterday, to that of today, or even that of tomorrow. Given a practical situation in which the term structure curve is continually changing in response to the arrival of new information, the use of any one of the above models as a practical tool for forecasting interest rate movements clearly becomes untenable.

The linking of current term structure rates to future rates is, however, made possible if one reformulates the above model in a state space framework. In order to do this one would need to define a set

$$\{(\tau_1, D(t, \tau_1)) \dots (\tau_p, D(t, \tau_p))\}$$

of p discount factor rate points, and corresponding term to maturity points, on the discount function curve currently prevailing in the market (see figure 3.1). Use can then be made of linear, quadratic or even cubic interpolation techniques to express in terms of $\{(\tau_1, D(t, \tau_1)) \dots (\tau_p, D(t, \tau_p))\}$ any other

point on the discount function curve. For example, if linear interpolation is to be used and $\tau_i < \delta < \tau_{i+1}$, then

$$D(t, \delta) = D(t, \tau_i) \frac{(\delta - \tau_{i+1})}{(\tau_i - \tau_{i+1})} + D(t, \tau_{i+1}) \frac{(\delta - \tau_i)}{(\tau_{i+1} - \tau_i)}$$

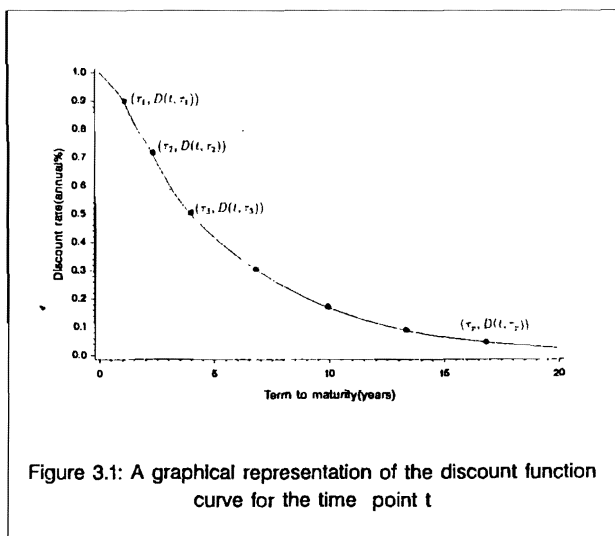


Figure 3.1: A graphical representation of the discount function curve for the time point t

while if quadratic interpolation is to be used and $\tau_i < \delta \leq \tau_{i+2}$ then

$$D(t, \delta) = D(t, \tau_i) \frac{(\delta - \tau_{i+1})(\delta - \tau_{i+2})}{(\tau_i - \tau_{i+1})(\tau_i - \tau_{i+2})} + D(t, \tau_{i+1}) \frac{(\delta - \tau_i)(\delta - \tau_{i+2})}{(\tau_{i+1} - \tau_i)(\tau_{i+1} - \tau_{i+2})} \\ + D(t, \tau_{i+2}) \frac{(\delta - \tau_i)(\delta - \tau_{i+1})}{(\tau_{i+2} - \tau_i)(\tau_{i+2} - \tau_{i+1})}$$

Substituting any one of the above expressions into (2.1) would then, in principle, yield a pricing model in which the only unknown parameters in the model are the p chosen discount factor rate points. Defining

$$\mathbf{x}_t = (D(t, \tau_1) \dots D(t, \tau_p))',$$

then, in order to complete the dynamic linear model formulation, all that one would need to do is specify a first-order Markov process of the form

$$\begin{aligned} \mathbf{x}_t &= \mathbf{G}\mathbf{x}_{t-1} + \mathbf{w}_t & \mathbf{w}_t &\sim N(0, \mathbf{Q}) \\ \mathbf{x}_0 &\sim N(\hat{\mathbf{x}}_{0|0}, \Sigma_{0|0}) \end{aligned}$$

for the evolution of the discount factor rate points, and to rewrite (3.1) so that the price of a coupon bearing stock is expressible as a linear function of $\mathbf{x}_t$.

As a practical illustration of how this can be done, let us assume that two discount factor rate points have been chosen to characterize the discount function curve and that a coupon bearing stock i that still has to make three coupon payments, and then, on maturing, a principal repayment of $N^{(i)}$ rand, is to be priced. Letting

$$\{\delta_1^{(i)}, \delta_2^{(i)}, \delta_3^{(i)}\}$$

denote the time remaining (in years), until each coupon payment is due then a direct application of (3.2.1) would yield

$$y_t^{(i)} = N^{(i)} C^{(i)} / 200 \sum_{j=1}^3 D(t, \delta_j^{(i)}) + N^{(i)} D(t, \delta_3^{(i)}) + e_t^{(i)}.$$

Assuming mainly for the purpose of illustration that

$$\mathbf{x}_t = (D(t, \delta_1^{(i)}), D(t, \delta_3^{(i)}))' = (x_t^{(1)}, x_t^{(2)})'$$

then, on making use of the following linear interpolation techniques, one could obtain

$$\begin{aligned} D(t, \delta_1^{(i)}) &= x_t^{(1)} \\ D(t, \delta_2^{(i)}) &= x_t^{(1)} + \frac{\delta_2^{(i)} - \delta_1^{(i)}}{\delta_3^{(i)} - \delta_1^{(i)}} (x_t^{(2)} - x_t^{(1)}) \\ D(t, \delta_3^{(i)}) &= x_t^{(2)} \end{aligned}$$

and thus rewrite the above pricing equation for stock i as

$$y_t^{(i)} = \left[N^{(i)} C^{(i)} / 200 - (N^{(i)} C^{(i)} / 200) \left[\frac{\delta_2^{(i)} - \delta_2^{(i)}}{\delta_3^{(i)} - \delta_1^{(i)}} \right] + N^{(i)} C^{(i)} / 200 \right] x_t^{(1)} \\ + \left[(N^{(i)} C^{(i)} / 200) \left[\frac{\delta_2^{(i)} - \delta_1^{(i)}}{\delta_3^{(i)} - \delta_1^{(i)}} \right] + N^{(i)} C^{(i)} / 200 + N^{(i)} \right] x_t^{(2)} + e_t^{(i)}.$$

The result is a dynamic linear stock pricing model of the form

$$y_t^{(i)} = f_t^{(i)'} x_t + e_t^{(i)} \\ x_t = G x_{t-1} + w_t$$

with

$$f_t^{(i)} = \begin{pmatrix} (N^{(i)} C^{(i)} / 200) \left[2 - \left[\frac{\delta_2^{(i)} - \delta_1^{(i)}}{\delta_3^{(i)} - \delta_1^{(i)}} \right] \right] \\ (N^{(i)} C^{(i)} / 200) \left[2 + \left[\frac{\delta_2^{(i)} - \delta_1^{(i)}}{\delta_3^{(i)} - \delta_1^{(i)}} \right] \right] \end{pmatrix}$$

and with

$$x_t = (x_t^{(1)}, x_t^{(2)})'$$

defining a discount factor rate vector whose elements pertain to a maturing period of $\delta_1^{(i)}$ and $\delta_2^{(i)}$ years respectively.

Turning our attention to the specification of a suitable model assumption for the random error vector v_t if we assume that

$$\text{var}(v_t) = \sigma^2 I_{n_t}$$

for all $t = 1, \dots, T$, and that

$$\text{cov}(v_t, v_s) = 0$$

for $t \neq s$, then a dynamic linear model formulation for the pricing of a sequence of fixed interest rate stock price observations can be completed as follows:

$$\begin{matrix} \mathbf{y}_t & = & \mathbf{F}_t & \mathbf{x}_t & + & \mathbf{v}_t & & \mathbf{v}_t \sim N(0, \sigma^2 \mathbf{I}_{n_t}) \\ n_t \times 1 & & n_t \times p & p \times 1 & & n_t \times 1 & & \end{matrix}$$

$$\begin{matrix} \mathbf{x}_t & = & \mathbf{G} & \mathbf{x}_{t-1} & + & \mathbf{w}_t & & \mathbf{w}_t \sim N(0, \mathbf{Q}) \\ p \times 1 & & p \times p & p \times 1 & & p \times 1 & & \end{matrix}$$

Flexibility in the forecasting ability of the above model can be enhanced if one allows the model parameters $\mathbf{G}$, $\mathbf{R}$ and $\mathbf{Q}$ to change in value over time in accordance with one's own views regarding the future course of movement in interest rate patterns. If, for example, the analyst feels that a fall of one percentage point in the shorter term rates is likely to occur he could decrease the state vector components that correspond to these spot rates accordingly, thereby modelling the term structure shift before it has actually occurred on the market. If, however, the analyst feels that the shorter term rates are going to change but is unsure as to the direction or magnitude of this change, he could choose rather to increase the entries in $\Sigma_{t|t}$ that correspond to these shorter term rates by an appropriate amount, thereby reflecting his uncertainty as to the future course of movement in these rates on the market. The above modelling approach thus provides a happy medium for incorporating subjective information into the model whilst at the same time allowing the model to update previous forecasts based on the information that is currently being provided by the stock price observations on the market.

3. AN APPLICATION USING WEEKLY STOCK PRICE DATA

A set of weekly stock price data was collected and analyzed over the time period 18/4/1986 to 26/9/1986. Employing the maximum likelihood estimation techniques that were developed earlier, and defining

$$\mathbf{x}_t = (x_t^{(1)}, \dots, x_t^{(6)})'$$

where

$$x_t^{(1)} = D(t, 0.5) = \text{the } \frac{1}{2}\text{-year discount rate}$$

$$x_t^{(2)} = D(t, 2) = \text{the 2-year discount rate}$$

$$x_t^{(3)} = D(t, 4) = \text{the 4-year discount rate}$$

$$x_t^{(4)} = D(t, 7) = \text{the 7-year discount rate}$$

$$x_t^{(5)} = D(t, 12) = \text{the 12-year discount rate}$$

$$x_t^{(6)} = D(t, 25) = \text{the 25-year discount rate}$$

the following results were obtained

$$\mathbf{G} = \begin{bmatrix} 0,4570853 & 0,3805243 & 0,2725211 & 0,1532933 & 0,0510388 & -0,01117 \\ 0,3795052 & 0,3181875 & 0,230287 & 0,1346883 & 0,0414934 & -0,006219 \\ 0,2717641 & 0,2277603 & 0,1745357 & 0,1091994 & 0,0316422 & -0,001564 \\ 0,152694 & 0,1304101 & 0,1111256 & 0,812717 & 0,0190636 & 0,0048186 \\ 0,05192 & 0,0421208 & 0,029846 & 0,0161396 & 0,0061568 & -0,002263 \\ -0,012464 & -0,00858 & -0,001545 & 0,003923 & -0,001134 & 0,0032925 \end{bmatrix}$$

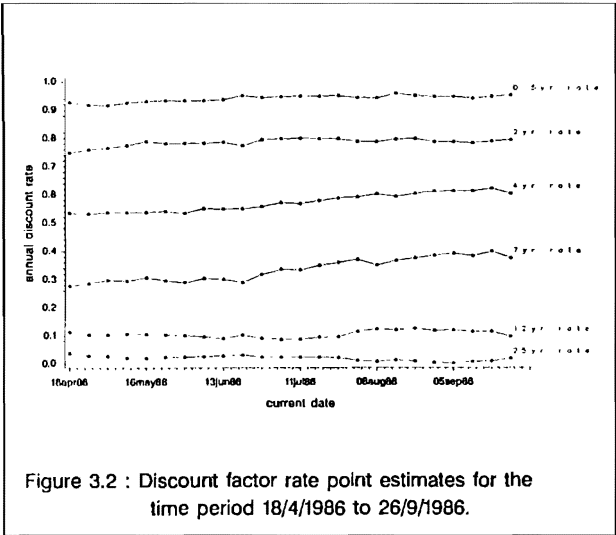
$$\mathbf{Q} = \begin{bmatrix} 0,0003068 & 0,0001942 & -0,000217 & -0,000469 & -0,000046 & -0,000121 \\ 0,0001942 & 0,0003402 & -0,000272 & -0,000415 & -0,000087 & -0,000103 \\ -0,000217 & -0,000272 & 0,0003975 & 0,0005201 & 0,0000807 & 0,0001227 \\ -0,000469 & -0,000415 & 0,0005201 & 0,001148 & 0,0000979 & 0,000262 \\ -0,000046 & -0,000087 & 0,0000807 & 0,0000979 & 0,000202 & 6,2795\text{E}-6 \\ -0,000121 & -0,000103 & 0,0001227 & 0,000262 & 6,2795\text{E}-6 & 0,0001679 \end{bmatrix}$$

$$\sigma^2 = 8,6896578$$

with parameter estimates for the above discount factor rates being given in figure 3.2. The associated variances for the above estimates are given in figure 3.3.

In order to compare the forecasting ability of our model with those of previously developed models, a term structure model similar to the one that was developed by Vasicek and Fong (1982) was computed for the time period date 18/4/1986. A naive assumption that no change would occur in the term structure rates over the observation period, 18/4/1986 to 26/9/1986, was then adopted from which a set of forecasts was developed. Letting $\tilde{e}_t = (\tilde{e}_t^{(1)}, \dots, \tilde{e}_t^{(n_t)})'$ denote the set of prediction error vectors that resulted from making such an assumption, a mean absolute prediction error criterion for the time period point t was defined as follows

$$MAPE = \frac{\sum_{i=1}^{n_t} |\tilde{e}_t^{(i)}|}{n_t}$$



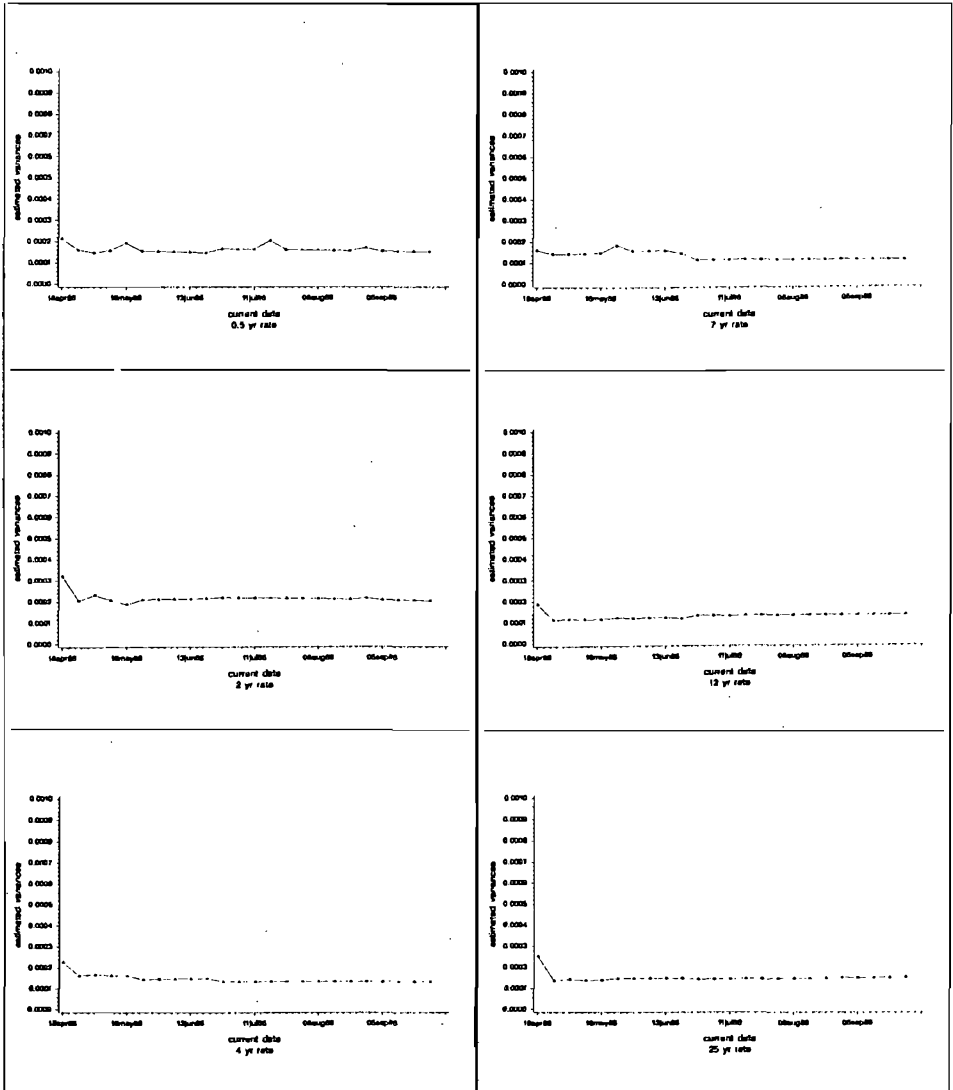
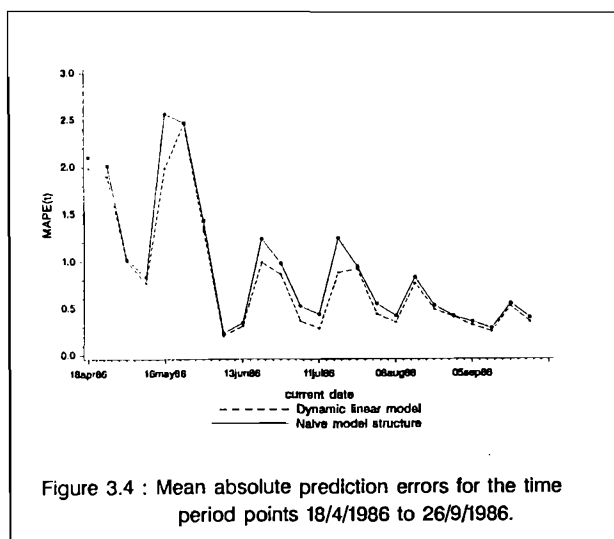


Figure 3.3 : Estimated variances for the chosen discount factor estimates.

Letting

$$MAPE = \frac{1}{n_t} \sum_{i=1}^{n_t} |e_t^{(i)}|$$

denote a corresponding mean absolute prediction error criterion for the dynamic linear model that was developed in this chapter, the plot in figure 3.4 clearly indicates a significant improvement in the forecasting performance of the dynamic linear model.



The choice of discount factor rates to include in the state vector were then varied, both in the number used, and their term to maturity. Although, with the use of a larger number of discount factor rate points, a very moderate improvement in the forecasting performance was observed, the corresponding increase in computing power that was required, did not, in our opinion, warrant the inclusion of more than six discount factor rate points in the state vector.

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